

# HARI PRASATH SM

Machine Learning Developer

+91 9489717803 | sm.hariprasath16@gmail.com | LinkedIn | Leetcode | Github | Portfolio

## SUMMARY

Artificial Intelligence & Data Science undergraduate (CGPA: 8.10) with proven experience in Deep Learning, NLP, and computer vision including a published patent in text-to-image generation. Proficient in Python, PyTorch, with 500+ competitive programming solutions on LeetCode. Eager to contribute to cutting-edge ML systems through an developer role.

## EDUCATION

### Sri Shakthi Institute of Engineering and Technology

B.Tech – Artificial Intelligence and Data Science

2023 – Present

CGPA: 8.10

### M.L.W.A. Hr. Sec. School

HSC – Maths & Computer Science

2022 – 2023

83.5%

## EXPERIENCE

### Machine Learning Intern

Cognifyz Technologies | Intern ID: CTI/A1/C106635

Feb 2025 – Mar 2025

- Analyzed manufacturer-provided machine dataset (timestamps, warranty details, change dates) using EDA to identify premature wear patterns in rubber belts and mechanical parts
- Identified root causes of accelerated machine degradation temperature anomalies, overloading, and substandard component quality through statistical analysis and feature correlation

## PROJECTS

### AgentScholar – Autonomous Multi-Agent Research System

2026

Python, LangGraph, LangChain, Ollama, Qwen 2.5 : 7B

- Built a 4-agent autonomous research pipeline (Planner, Searcher, Synthesizer, ReportWriter) orchestrated via LangGraph's StateGraph to conduct end-to-end research on any topic
- Implemented source credibility scoring (0–100) based on domain authority and HTTPS validation, filtering low-quality sources below configurable thresholds
- Generated structured citation-backed research reports in Markdown with APA/MLA/IEEE citation styles, with SQLite checkpointing for crash recovery and 7-day result caching

### VisionCrafter – Text-to-Image Generator

2024 – 2025

Python, PyTorch, BERT, GANs, COCO, OpenCV

- Built a deep learning pipeline combining BERT text encodings with GAN-based image synthesis, trained on the COCO dataset (100k+ image-caption pairs)
- Designed a custom GAN architecture conditioned on text embeddings to generate semantically accurate images from natural language prompts
- Filed patent (Application No. 202541052471, Published 27 June 2025) for the proposed text-to-image generation methodology

## SKILLS

**Languages:** Python, C, Dart, Flutter

**ML / DL:** TensorFlow, PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, OpenCV, Hugging Face, LangChain, LangGraph

**Tools & Platforms:** VS Code, PyCharm, Android Studio, Git, Google Colab, Kaggle, Docker

**Databases:** MySQL, Mongo DB, PostgreSQL

**Other:** OOP, DBMS, Problem Solving, REST APIs

## COMPETITIVE PROGRAMMING & HACKATHONS

- LeetCode: 530+ problems solved
- Smart India Hackathon 2025 – Built an AI-powered agricultural assistant for Kerala farmers featuring crop guidance, weather-aware advice, and pest management using React, FastAPI, and Groq LLM API (5-member team)
- VIT Hackathon (24hrs), Oct 2025 – Built a Medical RAG System with 92% retrieval accuracy as part of a 5-member team, Vellore Institute of Technology

## EXTRACURRICULAR ACTIVITIES

### Event Volunteer – Kalam 2025

2025

Sri Shakthi Institute of Engineering and Technology

- Organized and conducted coding events for the technical fest with 100+ participants

### Photographer & Video Editor – Media Guild

2024

Sri Shakthi Institute of Engineering and Technology

- Captured and edited photo/video content for technical and cultural college events

## CERTIFICATIONS

- Python – 100 Days of Code Python Bootcamp – Dr Angel Yu, Udemy
- Machine Learning – Complete Data Science, ML, DL, NLP Bootcamp – Krish Naik, Udemy

## ADDITIONAL INFORMATION

**Languages:** Tamil, English

**Areas of Interest:** Generative AI, LLMs, Computer Vision, Deep Learning, Open Source